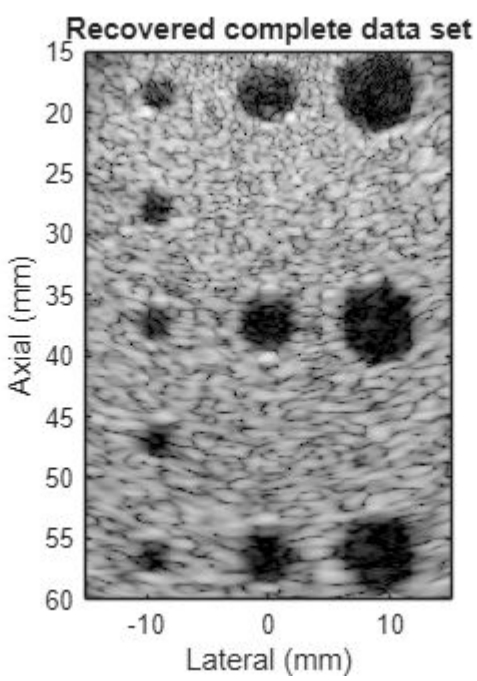
Demo

```
% Load the sample data and transmit delay timings
load('sample_data.mat');

% Optionally reduce the number of transmit events in the saved data
% ds=6;
% rf=rf(:,:,1:ds:end);
% transmit_delays=transmit_delays(1:ds:end,:);

% Perform decoding to produce the complete data set
rf_fsa=decode_focused_beams(rf,transmit_delays);

% Perform a basic diverging wave focusing of the complete data set
x=linspace(-15,15,200)/1000; % [m]
z=linspace(15,60,500)/1000; % [m]
r=(params.t0+(0:size(rf_fsa,1)-1))/params.fs*params.c; % [m] = c . tn
rf_focused=beamform(rf_fsa,r,params.rx_pos,x,z);

% Apply a high-pass filter to the data to remove interpolation artifacts
[b,a]=butter(2,500e3/(params.fs/2),'high');
rf_focused=filter(b,a,rf_focused);

% Display image
env=abs(hilbert(rf_focused));
env=env/max(env(:));
imagesc(x*1e3,z*1e3,db(env),[-50 0]);axis image;colormap gray
xlabel('Lateral (mm)')
ylabel('Axial (mm)')
title('Recovered complete data set')
```



```
% Load the saved figures
% openfig('dynamic_receive.fig')
% openfig('recovered_complete.fig')

% s**-1 / m*s^-1
```

Decode

Intuition in plain words

This function says:

- “I know those delays.”
- “So I can undo the coding caused by those delays.”
- “After undoing them, I can reorganize the data as if I had access to transmit-element-resolved data.”

The interpolation in the time version is simply the mechanism used to apply **fractional sample shifts**.

```
function [rf_fsa] = decode_focused_beams(rf,transmit_delays,method)
%DECODE_FOCUSED_BEAMS - Decode RF channel data from a set of focused
% transmit beams into the complete data set using the applied focal timings
%
% rf_fsa = decode_focused_beams(rf,transmit_delays);
% rf_fsa = decode_focused_beams(rf,transmit_delays, METHOD);
%
% Decoding can be performed in either the time or frequency domains. The
% two methods produce the same result, but frequency is faster.
%
% Parameters:
% rf - RF data (time sample x receive channel x transmit event)
% transmit_delays - Transmit focal delays in samples (transmit event x transmit
element)
% method - 'frequency' or 'time' (default: frequency)
%
% Returns:
% rf_fsa - Decoded complete data set (time sample x receive channel x transmit
element)
%
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% Check inputs
if(~exist('method','var'))
    method='frequency';
else
    assert(strcmp(method,'frequency')||strcmp(method,'time'),'Method must be
''frequency'' or ''time''')
end
```

```
[n_samples, n_receives, n_transmits]=size(rf);
n_elements=size(transmit_delays,2);
assert(size(transmit_delays,1)==n_transmits,'Transmit event dimensions inconsistent
between rf and transmit_timings')

switch method

    % ======================================Frequency domain implementation
    case 'frequency'

    % 1-D FFT to convert time to frequency
    RF=fft(single(rf));
    RF=permute(RF,[2 3 1]); % (receive channel x transmit event x time sample)
    frequency=(0:n_samples-1)/n_samples;

    % Apply decoding matrix at each frequency
    RF_adj=zeros(n_samples,n_receives,n_elements,'single');
    for i=1:ceil(n_samples/2) % only compute half, assume symmetry
        omega=2*pi*frequency(i);
        Hinv=exp(1j*omega*transmit_delays);
        RF_adj(i,:,:)=RF(:,:,i)*Hinv;
    end

    % Inverse FFT for real signal
    rf_fsa=ifft(RF_adj,'symmetric');

    % ==========================================Time domain implementation
    case 'time'

    rf_fsa=zeros(n_samples,n_receives,n_elements,'single');
    samples=1:n_samples;
    % Interpolate the data for each transmit event with delays
    % corresponding to each recovered transmit element
    for j=1:n_transmits
        rf_fsa=rf_fsa+interp1(samples,single(rf(:,:,j)),...
            repmat(samples(:),1,n_elements)
+repmat(transmit_delays(j,:),n_samples,1),...
            'linear');
    end
    rf_fsa(isnan(rf_fsa))=0;
end
end
```

Frequency-domain method: apply phase factors

$$Y(r,m,\omega) = \sum_{j=1}^{N_{\text{tx}}} X(r,j,\omega) e^{j\omega\tau_{j,m}}$$

- $X(r, j, \omega)$ is the RF from transmit event j
- $\tau_{j.m}$ is the delay applied to element m during event j

Then inverse FFT gives $y(r.m, n)$

-
- **Time-domain method**: shift/interpolate signals directly in time

Time-domain method: shift/interpolate signals directly in time

$q = k + a, 0 \leq a < 1$

then, $x(q) \approx (1 - \alpha)x(k) + \alpha\, x(k+1)$

`rf_fs(:,r,m) += interp( rf(:,r,j), samples +` $\tau_{j.m}$`)`

Each focused transmission is a coded combination of element emissions. So to recover element-based data, you combine information from all transmit events.

Beamforming

Intuition in plain words

This code says:

- "For each image pixel, I know how far it is from every element."
- "For each transmit-receive element pair, I know the total travel distance from TX to pixel to RX."
- "So I read the RF trace at that distance."
- "Then I sum all pair contributions."
- "The places where many pairs agree become bright."

That is beamforming.

`function [rf_focused] = beamform(rf_fsa,r,rx_pos,x,z)`

- `rf_fsa`: RF data indexed as**(range/time sample, receive element, transmit element)**
- `r`: the depth/range axis corresponding to the first dimension of `rf_fsa`
- `rx_pos`: positions of the array elements, probably one row per element, with coordinates like `[x y z]`
- `x`, `z`: coordinates of the image grid
- **`rf_focused`**: beamformed image on the `(z,x)` grid

```
% Perform a basic diverging wave beamforming on the complete data set
function [rf_focused] = beamform(rf_fsa,r,rx_pos,x,z)

% (1)
% Pre-calculate distances
[xg,zg]=meshgrid(x,z);
```

```
xg=xg(:);
zg=zg(:);

% (2)
delays=zeros(length(z)*length(x),size(rx_pos,1),'single');
for i=1:size(rx_pos,1)
    delays(:,i)=sqrt((xg-rx_pos(i,1)).^2+(zg-rx_pos(i,3)).^2);
end

% (3)
%Loop over element pairs and focus, sum
rf_focused=zeros(length(z)*length(x),1,'single');
for i=1:size(delays,2)
    for j=1:size(delays,2)
        rf_focused=rf_focused+interp1(r,rf_fsa(:,i,j),delays(:,i)+delays(:,j));
    end
end
rf_focused=reshape(rf_focused,length(z),length(x));
end
```

Big picture

For every image pixel (x,z), the code does this:

1. Compute the distance from that pixel to every array element
2. For every transmit-receive element pair (j,i):

   - take the RF signal `rf_fsa(:,i,j)`
   - sample it at the total path lengthdi+djd_i + d_jdi$_i$+dj$_j$
   - add that contribution to the pixel

3. Reshape the final vector into an image

So it is summing all transmit-receive pair contributions at the expected arrival time for each pixel.

(1) Build the image grid

```
% [xg,zg]=meshgrid(x,z);
% xg=xg(:);
% zg=zg(:);
```

(2) Compute pixel-element distances

```
% delays=zeros(length(z)*length(x),size(rx_pos,1),'single');
% for i=1:size(rx_pos,1)
%     delays(:,i)=sqrt((xg-rx_pos(i,1)).^2+(zg-rx_pos(i,3)).^2);
% end
```

For element i at x_i, z_i

$$d_i(x_p, z_p) = \sqrt{(x_p - x_i)^2 + (z_p - z_i)^2}$$

- Rows of delays correspond to pixels
- Size of delays is $(N_z N_x) \times N_{\text{elem}}$
- So `delays(k,i) is distance from pixel "k" to element number "i"`
- `This delays are measure in distance, [m] not time`

(3) Beamform image loop over Rx/Tx element pairs

Initialization based on grid space z, x

`rf_fsa(:,i,j)`

`Transmit element "j" and receive element "i"`

For a pixel at position (x,z)(x,z)(x,z), if transmit element `j` emits and receive element `i` records, then the total propagation path is: $d_{\text{tx},j}(x, z) + d_{\text{rx},i}(x, z)$

Since the code uses the same `rx_pos` array for both transmit and receive geometry, it assumes the same array positions for TX and RX.

Thus:

- `delays(:,j)` = distance from pixel to transmit element `j`
- `delays(:,i)` = distance from pixel to receive element `i`

Total path is $d_j + d_i$

Interpolation explanation

Because the total path:di+djd_i+d_jdi$\hat{}$+dj$\hat{}$

usually does **not** fall exactly on one value of `r`, `interp1` estimates the signal between samples.

For example, if:

- `r = [0, 0.1, 0.2, 0.3, ...]`
- query point is `0.26`
- `Basically if s(0.26) = 0.4 s(0.2) + 0.6 s(0.3)`

then MATLAB interpolates between the samples at `0.2` and `0.3`.

$$I(x_p, z_p) = \sum_{i=1}^{N} \sum_{j=1}^{N} s_{\text{ij}}(d_i(x_p, z_p) + d_j(x_p, z_p))$$

where:

- $s_{\mathrm{ij}}(r)$ is `rf_fsa(:,i,j)`
- d_i distance from pixel to element i
- d_j distance from pixel to element j